MILITARY SPECIFICATION

BURNER, GAS, LABORATORY AND CARTRIDGE, ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a laboratory gas burner and cartridge assembly which is suitable as a portable gas heating source for laboratory use.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

MILITARY

MIL-P-116	- Methods of Preservation-packaging.
MIL-L-10547	- Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

(DD Form 1426 will be provided in final document.)

FSC 6640

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, and drawing required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

HAZARDOUS MATERIALS REGULATIONS OF THE DEPARTMENT OF TRANSPORTATION.

R. M. Graziano's Tariff No. 29 Publishing Hazardous Materials Regulations of the Department of Transportation.

(Application for copies should be addressed to the Bureau of Explosives, Association of American Railroads, 1920 L St., Washington, DC 20036).

3. REQUIREMENTS

3.1 Material.

3.1.1 Burner. All components, except the manual valve adjustment parts, shall be made of corrosion-resisting steel, brass or aluminum. The manual valve adjustment knob shall be made of a thermosetting plastic.

3.1.2 Cartridge.

3.1.2.1 Tank. The tank shall be made of good quality commercial metal, suitably treated to resist corrosion, and suitable for containing liquefied burner fuel under pressure.

3.1.2.2 Fuel. The fuel shall be a nontoxic liquid blend of petroleum products.

3.2 Construction and design. The laboratory gas burner and cartridge assembly shall consist of a laboratory gas burner and a laboratory gas burner cartridge as shown in figure 1 and as described in 3.2.1 and 3.2.2.

3.2.1 Laboratory gas burner. The laboratory gas burner shall be constructed in accordance with figure 1 and shall consist of a valve assembly, jet tube, baffle, air-adjusting ring, burner insert and burner tube. Laboratory gas burner shall be 4-1/2 ± 1/4 inches long.

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3.2.1.1 Valve assembly. The valve assembly shall be fitted with a manual adjustment knob to turn on or turn off the fuel supply, and shall have an internal (7/16-27 NS-2) thread at the distal end for coupling to the laboratory gas burner cartridge (see 3.2.2).

3.2.2 Laboratory gas burner cartridge. The laboratory gas burner cartridge shall be constructed in accordance with figure 1 and shall consist of a cylindrical, metal, disposable, nonrefillable tank containing a nominal 170 grams by weight of liquefied petroleum fuel.

3.2.2.1 Tank. The top of the fuel tank shall be fitted with a spring-loaded, ball-type, brass valve with an external thread suitable for coupling to the laboratory gas burner (see 3.2.1).

3.3 Performance.

3.3.1 Constant flow. When tested as specified in 4.4.1, the laboratory gas burner and cartridge assembly shall exhibit a constant flame size at each flow position of the manual adjustment valve.

3.3.2 Flame length. When tested as specified in 4.4.2, the laboratory gas burner and cartridge assembly air-adjusting ring shall limit the maximum attainable length of blue flame to six inches.

3.4 Identification marking.

3.4.1 Laboratory gas burner. Each burner shall be legibly and permanently marked with the name or registered trademark of the manufacturer.

3.4.2 Laboratory gas burner cartridge. Each fuel cartridge shall be affixed with a label bearing the following data:

a. The item name - "Cartridge, Laboratory Gas Burner."
b. Name and address of manufacturer.
c. Instructions for use.
d. Net weight of contents.
e. Precautionary statements in regard to flammability, safety and handling.

3.5 Workmanship. Workmanship shall be first class throughout. The burners and cartridges shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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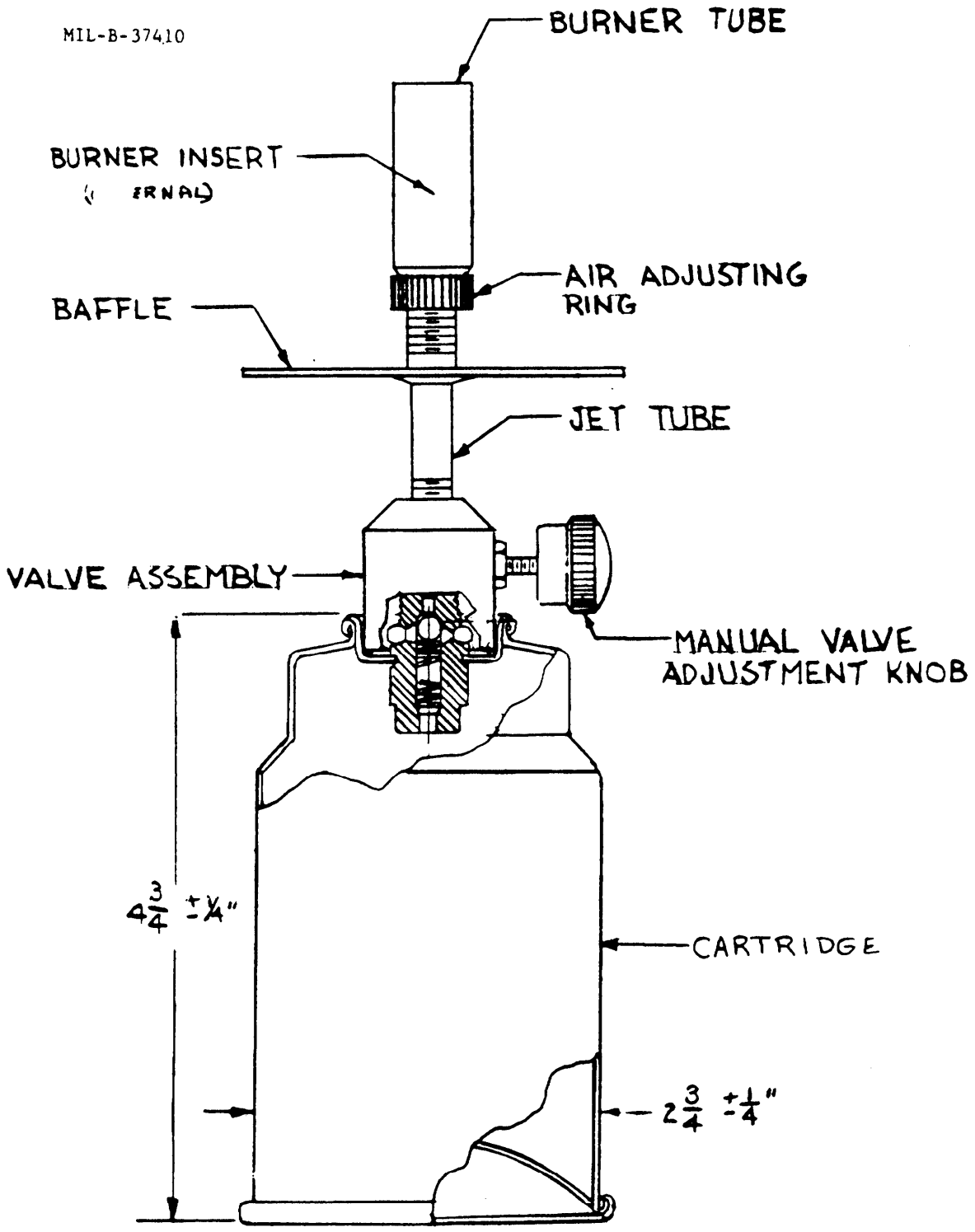


FIGURE 1. Burner, gas, laboratory and cartridge assembly

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I.

TABLE I. For examination.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	4.0
For dimensional examination	S-3	2.5

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II.

TABLE II. Tests

Component (Unit of Product)	Characteristic	Requirement	Test procedure	Inspection level	AQL
Burner and cartridge assembly	Constant flow	3.3.1	4.4.1	S-1	*
Burner	Constant flow	3.3.1	4.4.1	S-1	*
Burner cartridge	Constant flow	3.3.1	4.4.1	S-1	*
Burner and cartridge assembly	Flame length test	3.3.2	4.4.2	S-1	*
Burner	Flame length test	3.3.2	4.4.2	S-1	*
Burner cartridge	Flame length test	3.3.2	4.4.2	S-1	*

* AQL shall be zero.

4.3 Examination. The burner and cartridge assembly shall be examined to determine compliance with all requirements contained in this specification.

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4.3.1 Classification of defects. Examination shall be conducted in accordance with table III. Item shall be examined for defects including but not limited to the following:

TABLE III. Classification of defects.

Major:	
101	Component missing.
102	Components not free from pits, burrs or scratches.
103	Burner manual valve adjustment parts cannot be tightly secured to prevent fuel leakage.
104	Burner air-adjusting ring not adjustable for complete air control.
105	Burner does not produce a hot clean flame.
106	Burner cannot be securely attached to fuel cartridge.
107	Top of fuel tank not sufficiently crimped to prevent fuel leakage.
Minor:	
201	Burner or cartridge not free from dirt, grease and foreign particles.
202	Identification marking not present, complete, correct, legible and permanent.

4.3.2 Dimensional examination. The burner and cartridge assembly shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

4.3.3 Packaging inspection. The inspection of the packaging and interior package marking shall be in accordance with the group A quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Constant flow test. The manual adjustment on the valve assembly of the lighted laboratory gas burner and cartridge assembly shall be adjusted through the entire range producing gas flow.

4.4.2 Flame length test. The air-adjusting ring of the laboratory gas burner and cartridge assembly shall be adjusted through the entire blue flame range.

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5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit package. Each burner and cartridge as specified, shall be individually packaged in accordance with MIL-P-116, method III in a box conforming to PPP-B-566, PPP-B-676 or PPP-B-636, type CF, class domestic. Closure of box shall be as specified in the appendix of the applicable box specification.

5.1.1.2 Intermediate packaging. Six (6) cartridges shall be packaged in a box of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic. Closure of box shall be as specified in the appendix of PPP-B-636. Intermediate package is not required for the burners.

5.1.2 Level C. Burners or cartridges shall be individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Twenty-four (24) burners or thirty-six (36) cartridges shall be packed in an exterior container designed for a type 2 load constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproof as specified in the appendix of PPP-B-636.

5.2.2 Level B. Twenty-four (24) burners or thirty-six (36) cartridges shall be packed in an exterior container designed for a type 2 load constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The packaged burners or cartridges shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carriers rules and regulations.

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5.3 Marking. Each unit, intermediate package and exterior container shall be marked as specified in MIL-STD-129. In addition, the cartridges shall be marked as required by the Hazardous Material Regulations of the Department of Transportation.

6. NOTES

6.1 Intended use. This item provides a portable gas energy source for general laboratory use.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 National Stock Number (NSN) coverage. This specification is applicable to the following items listed in the Federal Supply Catalog:

National Stock Number	Item Identification
6640-00-933-8868	BURNER, GAS, LABORATORY, 4-1/2 inches Long.
6640-00-926-7674	CARTRIDGE, LABORATORY GAS BURNER, Disposable, 170 Gm.

6.4 Limitations. This specification does not cover all types of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover those items which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DSA - DM

Project No. 6640-0973.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE
MIL-B-37410 BURNER, GAS, LABORATORY AND CARTRIDGE, ASSEMBLY

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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